VARDHAMAN COLLEGE OF ENGINEERING

Department of Electronics and Communication Engineering

ePBL INTERNSHIP COURSE-END PROJECT REPORT

RTL-TO-GDSII PHYSICAL DESIGN AND IMPLEMENTATION OF A 4-BIT SEQUENTIAL ADDER USING SYNOPSYS ICC2

Course: VLSI Design and Systems

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Academic Year 2025-2026

CERTIFICATE

This is to certify that the ePBL Internship course-end project report entitled "RTL-to-GDSII Physical Design and Implementation of a 4-bit Sequential Adder Using Synopsys ICC2" is a record of the work carried out by Suraj Kuila (23881A04C0), Naradasu Srija (23881A04A9), Patniti Vaishnavi (23881A04B3), and Tirunagari Akshay (23881A04C3) under the guidance of Dr. K. Narsimha Reddy for the course VLSI Design and Systems at Vardhaman College of Engineering during the academic year 2025-2026.

The work presented in this report documents the implementation of the major ASIC backend stages, including technology setup, design import, floorplanning, power planning, placement, clock tree synthesis, routing, filler-cell insertion, report analysis, and GDSII generation.

Project Coordinator **Dr. K. Narsimha Reddy**	Head of the Department

DECLARATION

We hereby declare that the project work presented in this report is an original record of the implementation and analysis carried out by our team for the ePBL Internship Program. The report has been prepared from the supplied design reports, implementation scripts, and ICC2 layout screenshots. All machine-specific paths and unrelated institutional references have been removed from the documented flow. The work has not been submitted elsewhere for the award of another academic qualification.

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We also thank the Department of Electronics and Communication Engineering, Vardhaman College of Engineering, for providing the academic environment and access required to study an industry-standard RTL-to-GDSII implementation methodology. Finally, we thank all the members of the project team for their cooperation in executing the flow, collecting reports, analyzing results, and preparing this report.

ABSTRACT

This project presents the physical design and implementation of a 4-bit sequential adder using Synopsys IC Compiler II (ICC2). The design contains registered operand and result paths, making it suitable for studying clock-aware physical implementation and register-to-register timing. The implementation flow begins with the import of a synthesized gate-level netlist and SAED 32 nm technology data. High-threshold-voltage and low-threshold-voltage standard-cell libraries are linked, maximum and minimum TLU+ parasitic models are loaded, and a functional slow-corner scenario at 0.75 V and 125 degrees Celsius is configured.

The design is then processed through floorplanning, pin placement, power-ground rail creation, timing-driven placement, clock tree synthesis, global routing, track assignment, detailed routing, filler-cell insertion, report generation, and GDSII export. The final synthesized report contains 27 cells, including 13 combinational cells and 14 sequential cells, with a total reported area of 147.661879 area units. The corresponding reported dynamic power is 6.4757 uW and leakage power is 2.0678 uW. These values show a substantial reduction relative to an earlier implementation state, although the final power report did not include a defined clock and must therefore be interpreted as an estimate rather than a complete signoff value.

The physical layout screenshots confirm that cell placement, routing, post-route visualization, filler insertion, and GDSII generation were exercised. Verification reports show that power-grid geometrical DRC completed without errors; however, the uploaded power-grid connectivity report identified floating VDD/VSS wires and floating standard cells. Similarly, the final synthesis timing path was unconstrained because the supplied SDC contained only unit definitions. The report therefore documents both the successful implementation stages and the remaining signoff actions required for a fully constrained, connectivity-clean final result.

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CHAPTER 1

INTRODUCTION

1.1 Background

Application-specific integrated circuit design is commonly divided into frontend and backend stages. The frontend stages describe and verify the logical behavior of a circuit and synthesize it into a technology-mapped gate-level netlist. The backend, or physical-design, stages transform that netlist into a geometrically legal and timing-aware layout suitable for final mask-data generation. Physical design is therefore the bridge between logical functionality and semiconductor fabrication.

A successful backend flow must simultaneously satisfy several interacting constraints. High placement density can reduce area but may increase congestion. Large buffers can improve transition and setup timing but increase area and power. Clock-tree construction improves clock distribution but introduces latency, skew, routing demand, and additional power consumption. Power-grid design must satisfy both geometrical design rules and electrical connectivity. These trade-offs make physical implementation an iterative engineering process rather than a single command execution.

1.2 Project Motivation

The purpose of this ePBL Internship project is to obtain practical exposure to the complete ASIC physical-design flow using an industry-standard implementation environment. A 4-bit sequential adder was selected because it contains both combinational arithmetic logic and clocked storage elements. This allows the project to demonstrate cell placement, clock-tree synthesis, register-to-register timing, routing, and physical verification within a design that remains small enough for detailed visual inspection and report analysis.

1.3 Problem Statement

The project problem is to convert a synthesized gate-level representation of a 4-bit sequential adder into a routed physical layout using Synopsys ICC2. The implementation must establish a valid technology environment, define the operating scenario, create a floorplan, distribute power and ground, place the standard cells, construct and route the clock network, route the signal nets, insert filler cells, generate implementation reports, and export the final design in GDSII format.

1.4 Objectives

1. Understand the complete RTL-to-GDSII backend implementation sequence.
2. Configure the SAED 32 nm technology, timing libraries, reference libraries, and TLU+ parasitic models.
3. Import and link the synthesized sequential-adder netlist in ICC2.
4. Define a functional slow-corner scenario at 0.75 V and 125 degrees Celsius.
5. Create a floorplan with 75% core utilization and suitable boundary offset.
6. Place input and output pins on controlled routing layers and chip sides.

7. Create VDD and VSS nets and standard-cell power rails.
8. Perform timing-driven placement and cell legalization.
9. Build and route a clock tree for the sequential elements.
10. Perform global routing, track assignment, and detailed routing.
11. Insert standard-cell fillers and export the final GDSII database.
12. Analyze area, power, timing, floorplan, and power-grid verification reports.
13. Identify incomplete constraints and connectivity issues that must be corrected before final signoff.

1.5 Scope and Limitations

The report covers the ASIC backend flow and the analysis of the uploaded synthesis and ICC2 artifacts. It does not include proprietary technology-library contents, licensed tool installation files, or fabrication data. Functional simulation waveforms and a complete post-route signoff timing report were not included in the uploaded artifacts. Consequently, functional correctness is assumed from the synthesized netlist, while timing closure and power-grid closure are evaluated only from the available reports.

1.6 Project Team

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Table 1.1: Project team and course information

Project Coordinator: Dr. K. Narsimha Reddy. Course: VLSI Design and Systems. Institution: Vardhaman College of Engineering.

CHAPTER 2

DESIGN SPECIFICATION AND TECHNOLOGY SETUP

2.1 Design Overview

The implemented block is named full_adder in the reports; however, the design statistics and layout labels show that it is a multi-bit registered arithmetic block rather than a single combinational one-bit full adder. The design contains 14 sequential cells and labels such as regA, regB, SUM_reg, C_out_reg, and t_reg. It is therefore documented in this report as a 4-bit sequential or registered adder. The synthesized block has 15 ports, 27 cells in the optimized report, 13 combinational cells, and 14 sequential cells.

2.2 Functional Architecture

The architecture stores operand data in input registers, performs combinational addition, and captures the result in output registers. The visual implementation shows four-bit register groups for the two operands and the sum, together with a registered carry output and supporting control storage. This organization creates clock-to-Q, combinational, setup, and hold timing paths that can be analyzed during physical implementation.

2.3 Technology and Library Configuration

Parameter	Configuration
Technology node	SAED 32 nm
Metal stack	1P9M technology file
Reference libraries	High-Vt and Low-Vt NDM standard-cell libraries
Timing corner	Slow-slow, 0.75 V, 125 degrees Celsius
Power supply	VDD = 0.75 V; VSS = 0 V
Late parasitic model	Maximum TLU+ model
Early parasitic model	Minimum TLU+ model
Operating mode	Functional
Scenario	Functional slow-corner maximum-capacitance scenario
Implementation tool	Synopsys IC Compiler II
Report tool version	T-2022.03-SP5

Table 2.1: Technology, library, and operating-corner configuration

2.4 Parasitic Technology Models

Interconnect resistance and capacitance are represented using TLU+ parasitic technology files. The maximum model is assigned to late analysis and the minimum model to early analysis. This distinction is important because maximum interconnect delay is normally relevant to setup analysis, while minimum delay is relevant to hold analysis. The layer-map file establishes the correspondence between the technology-file layer names and the parasitic-model layer names.

2.5 Design Statistics

Metric	Optimized reported value
Ports	15
Nets	38
Total cells	27
Combinational cells	13
Sequential cells	14
Buffers/Inverters	1
Macros or black boxes	0
Cell references	5

Table 2.2: Synthesized design statistics

2.6 Timing Constraints

A 10 ns clock was created in the ICC2 physical-design script for the port named Clock. This corresponds to a target clock frequency of 100 MHz. However, the uploaded full_adder.sdc file contains only unit definitions and does not contain a create_clock command, input delays, output delays, uncertainty, or electrical constraints. This explains why one uploaded timing report marks the final output path as unconstrained and why one power report warns that no clock is defined.

Constraint interpretation

The 10 ns clock used during ICC2 implementation should be added directly to the final SDC, together with input delay, output delay, clock uncertainty, transition, load, and any false-path or multicycle-path exceptions. A timing report is not considered closed when the reported path is unconstrained.

CHAPTER 3

ASIC PHYSICAL DESIGN METHODOLOGY

3.1 Overall Flow

Stage	Physical design operation
1	Technology and library setup
↓	
2	Gate-level netlist import and linking
↓	
3	MMMC scenario and timing-constraint setup
↓	
4	Floorplanning and IO pin placement
↓	
5	Power and ground planning
↓	
6	Placement and pre-CTS optimization
↓	
7	Clock Tree Synthesis and clock optimization
↓	
8	Global, track, and detailed routing
↓	
9	Filler insertion and final checks
↓	
10	GDSII generation

Figure 3.1: Standard RTL-to-GDSII physical design flow

3.2 Stage Objectives

Stage	Primary objective
Library setup	Create the design library and load physical, timing, and parasitic technology data.
Design import	Read the synthesized netlist, set the top module, and resolve reference cells.
MMMC setup	Create the functional mode, PVT corner, scenario, and timing constraints.
Floorplanning	Define die/core geometry, utilization, rows, offsets, and pin

Stage	Primary objective
	locations.
Power planning	Create VDD/VSS nets and connect standard-cell power rails.
Placement	Place and legalize cells while reducing congestion and improving timing.
CTS	Distribute the clock with controlled skew, latency, and transition.
Routing	Connect signal nets through legal metal tracks and vias.
Fill and signoff	Fill placement gaps, check routes and PG, and generate final reports.
GDSII export	Write the final physical layout database.

Table 3.1: Physical design stages and objectives

3.3 Floorplan Strategy

The command history includes experiments with U-shaped or L-shaped geometry and a rectangular floorplan. Executing two initialize_floorplan commands in sequence would overwrite the first geometry. For a reproducible final flow, the rectangular floorplan is retained with 0.75 core utilization, a 1:1 side ratio, and a core offset of 10. The L-shaped screenshots are preserved in the report as evidence of design-planning exploration.

3.4 Pin-Placement Strategy

Early commands constrained input pins to M1 and output pins to M2 with 5-unit spacing. The final script visible in the uploaded screenshot uses M3 for input pins on sides 1 and 3, M4 for output pins on sides 2 and 4, and a pin spacing of 7. Moving block pins to higher routing layers can reduce local access competition near standard-cell rails. The final report therefore records M3/M4 as the last documented pin-placement configuration.

3.5 Power-Planning Strategy

The implementation creates VDD and VSS logical nets, connects them to hierarchical cell power pins, defines a standard-cell rail pattern on M1, applies the pattern to the core through a power-grid strategy, and compiles the strategy. The sequence is then checked using power-grid connectivity, open, and DRC commands. Geometry and connectivity are separate requirements: a power structure can satisfy width and spacing rules while still remaining electrically disconnected.

3.6 Placement and Clock Strategy

Placement is executed in explicit ICC2 stages, progressing from initial placement through design-rule repair, initial optimization, final placement, and final optimization. After placement, a 10 ns clock is defined and clock_opt is run through build_clock, route_clock, and final_opto stages. The clock network is required because the design contains 14 sequential cells.

3.7 Routing and Finishing Strategy

Signal routing is divided into route_global, route_track, and route_detail operations. After routing, filler cells are inserted to occupy unused placement sites, maintain row continuity, and support well and implant continuity. The final layout is written to GDSII. A binary GDS file should be inspected in a layout viewer rather than opened in a normal text editor.

CHAPTER 4

IMPLEMENTATION PROCEDURE AND SCRIPTS

4.1 Project Directory Preparation

A reproducible implementation separates design inputs, scripts, reports, logs, saved blocks, and final results. The commands file mkdir ./results and file mkdir ./reports create the primary output directories. A complete project may use the following organization.

```
project/
├── inputs/
│   ├── design_netlist.v
│   └── design_constraints.sdc
├── scripts/
├── reports/
├── results/
├── logs/
└── run_all.tcl
```

4.2 Library and Parasitic Setup

The design library is created from the SAED 32 nm technology file and HVT/LVT NDM reference libraries. Timing libraries are assigned for linking and target-cell selection. Maximum and minimum TLU+ models are then loaded.

```
set TECH_FILE       "/path/to/technology.tf"
set REF_LIBS        [list "/path/to/hvt.ndm" "/path/to/lvt.ndm"]
set DB_LIBS         [list "/path/to/hvt_slow.db" "/path/to/lvt_slow.db"]

create_lib -technology $TECH_FILE -ref_libs $REF_LIBS $DESIGN_LIBRARY
set_app_var link_library   [concat {*} $DB_LIBS]
set_app_var target_library $DB_LIBS

read_parasitic_tech -name maxTLU -tlup $MAX_TLUPLUS -layermap $TLU_MAP
read_parasitic_tech -name minTLU -tlup $MIN_TLUPLUS -layermap $TLU_MAP
```

4.3 Netlist Import

```
read_verilog $NETLIST_FILE     -library $DESIGN_LIBRARY     -design "${DESIGN_NAME}_net"     -top
$DESIGN_NAME

current_design $DESIGN_NAME
link_block
check_design
```

4.4 Mode, Corner, Scenario, and SDC

```
remove_scenarios -all
remove_modes -all
remove_corners -all

create_mode functional
create_corner slow_max
set_voltage -object_list {VDD} 0.75
set_voltage -object_list {VSS} 0.0
set_temperature -corners slow_max 125
set_parasitic_parameters -corners slow_max     -late_spec maxTLU -early_spec minTLU
```

```
create_scenario -name functional_slow_max     -mode functional -corner slow_max
current_scenario functional_slow_max
read_sdc $SDC_FILE
```

4.5 Floorplanning and Pin Placement

```
initialize_floorplan     -shape R     -core_utilization 0.75     -side_ratio {1 1}     -core_offset 10

set_block_pin_constraints     -allowed_layers M3 -sides {1 3} -pin_spacing 7 -self
place_pins -ports [all_inputs]

set_block_pin_constraints     -allowed_layers M4 -sides {2 4} -pin_spacing 7 -self
place_pins -ports [all_outputs]
```

4.6 Power Planning

```
create_net -power VDD
create_net -ground VSS
connect_pg_net -net VDD [get_pins -hierarchical "*/VDD"]
connect_pg_net -net VSS [get_pins -hierarchical "*/VSS"]

create_pg_std_cell_conn_pattern std_rail_pattern -layers {M1}
set_pg_strategy rail_strategy -core     -pattern {{name: std_rail_pattern} {nets: VDD VSS}}
compile_pg -strategies rail_strategy

check_pg_connectivity
check_pg_opens
check_pg_drc
```

4.7 Placement

```
place_opt -from initial_place -to initial_place
place_opt -from initial_drc   -to initial_drc
place_opt -from initial_opto  -to initial_opto
place_opt -from final_place   -to final_place
place_opt -from final_opto    -to final_opto
save_block -as post_placement
```

4.8 Clock Tree Synthesis

```
create_clock -name clk -period 10 [get_ports Clock]
clock_opt -from build_clock -to build_clock
clock_opt -from route_clock -to route_clock
clock_opt -from final_opto  -to final_opto
report_timing
save_block -as post_cts
```

4.9 Routing, Fillers, and GDSII

```
route_global
route_track
route_detail
save_block -as post_route

set FILLER_CELLS [get_lib_cells -quiet */*FILL*]
create_stdcell_fillers -lib_cells $FILLER_CELLS

check_routes
check_pg_connectivity
check_pg_drc
write_gds ./results/final_layout.gds
save_block -as post_gds
```

Command	Purpose
create_lib	Creates the ICC2 design library with technology and reference libraries.
read_parasitic_tech	Loads TLU+ resistance/capacitance models.
read_verilog	Imports the synthesized gate-level netlist.
create_mode/create_corner	Defines the MMMC operating environment.
initialize_floorplan	Creates die/core geometry and placement rows.
place_pins	Places input and output ports.
compile_pg	Compiles the standard-cell power-grid strategy.
place_opt	Performs placement, legalization, and optimization.
clock_opt	Builds, routes, and optimizes the clock tree.
route_global/track/detail	Completes signal routing.
create_stdcell_fillers	Inserts filler cells in unused row sites.
write_gds	Exports the final GDSII layout database.

Table 4.1: Major commands used in the ICC2 flow

CHAPTER 5

PHYSICAL IMPLEMENTATION EVIDENCE

5.1 Initial Design Import and Floorplan

The initial ICC2 view shows the imported standard-cell instances positioned inside the generated core and the block pins distributed around the outer boundary. Sequential cells are visible as DFFX1_HVT instances, while XOR/XNOR and complex logic cells implement the arithmetic function.

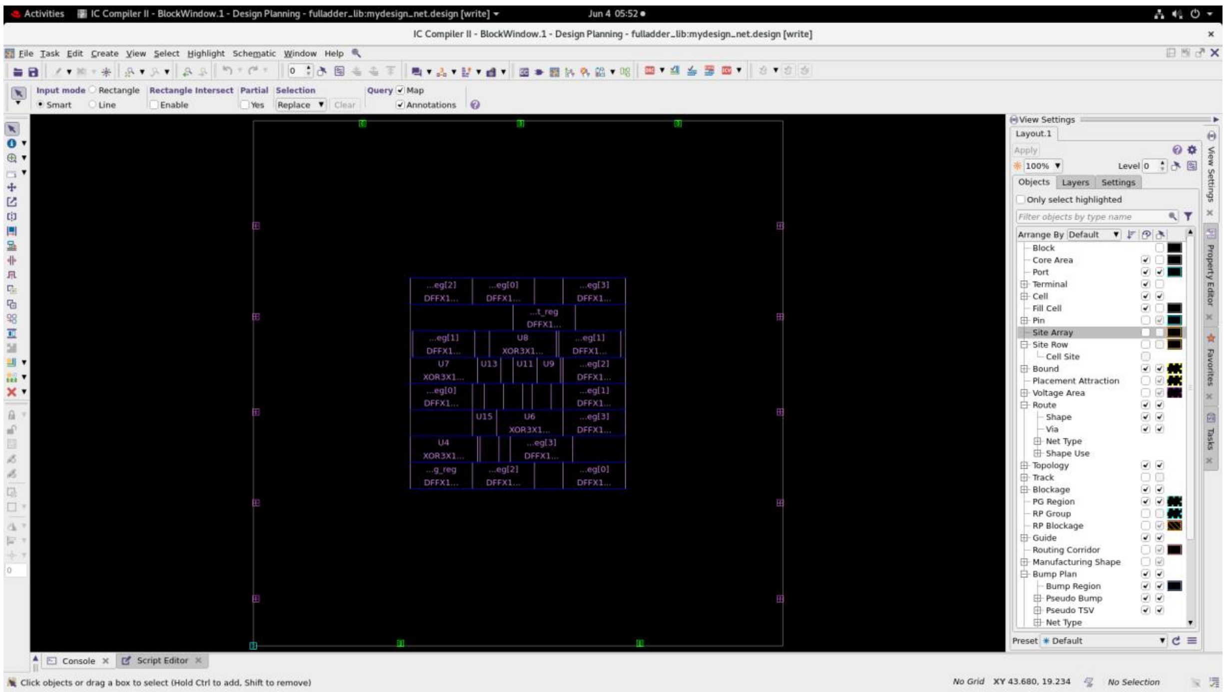

Figure 5.1: Initial design view after import and floorplan creation

5.2 Standard-Cell Placement

The placed design shows standard cells organized in legal rows within the core. The central placement region contains operand registers, output registers, carry storage, and combinational adder logic. The large whitespace outside the core reflects the chosen die/core offset and the small size of the training design.

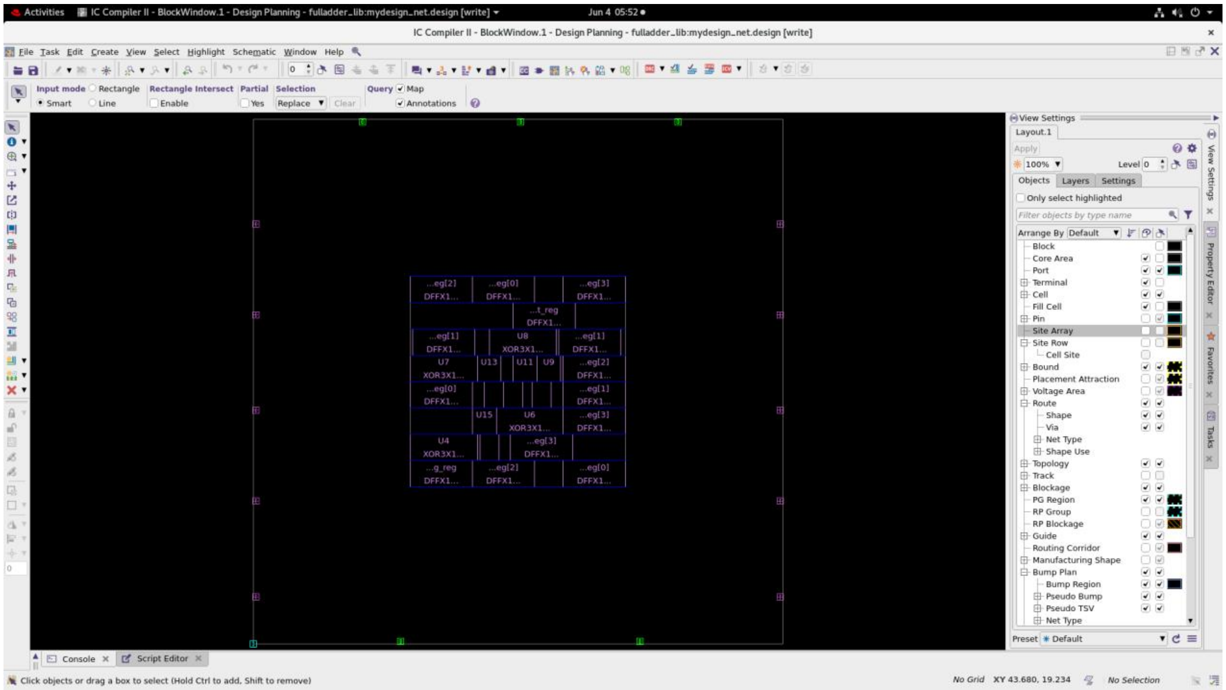

Figure 5.2: Standard-cell placement inside the core region

5.3 L-shaped Floorplan Exploration

Two screenshots document an L-shaped or non-rectangular floorplan experiment. Such a floorplan can be useful when integrating around fixed macros or blocked regions, but it creates irregular row lengths and may complicate routing. Since this design contains no macros, the final documented flow uses a rectangular floorplan for reproducibility.

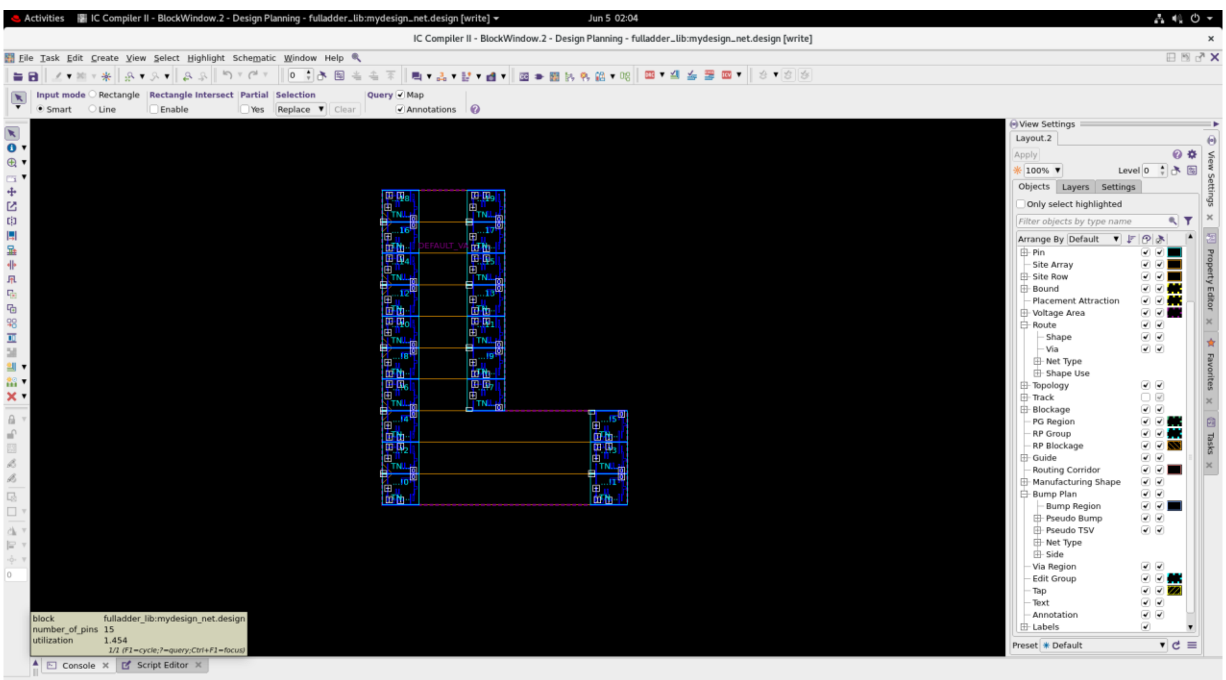

Figure 5.3: Experimental L-shaped floorplan view

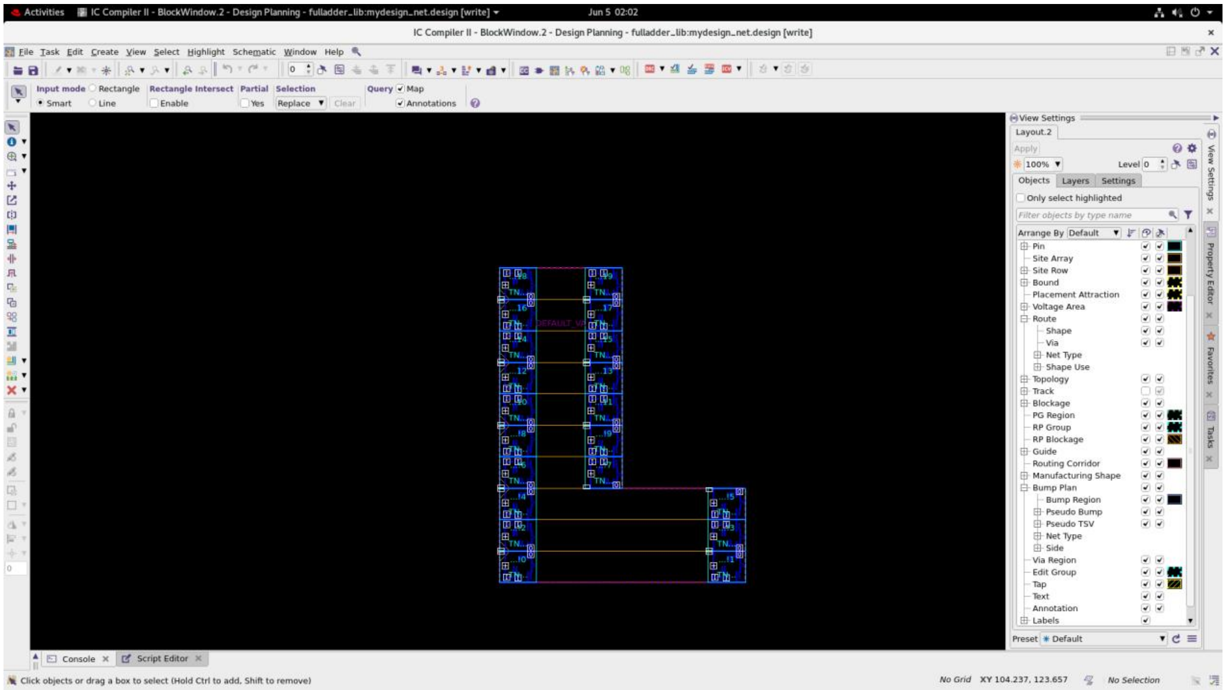

Figure 5.4: L-shaped layout with cell rows and connectivity

5.4 Routing

The routing screenshot shows the placement rows, standard cells, and routed nets after global, track, and detailed routing. The visible horizontal and vertical interconnects connect operand registers, combinational logic, sum registers, carry logic, and external ports.

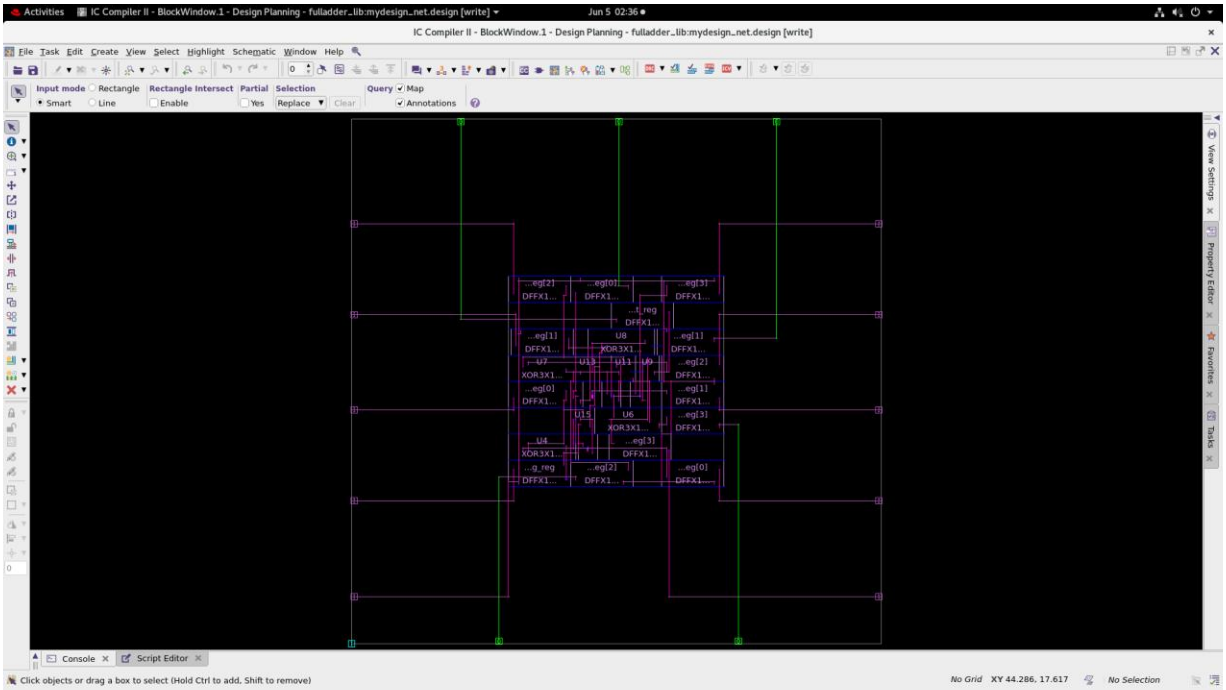

Figure 5.5: Routed layout visualization

5.5 Post-route Layout

The post-route view provides a closer inspection of metal connections and cell locations. The design contains a compact central logic region with long routes extending to pins on the block boundary. These long boundary routes illustrate why pin placement and die size influence wirelength and parasitic delay.

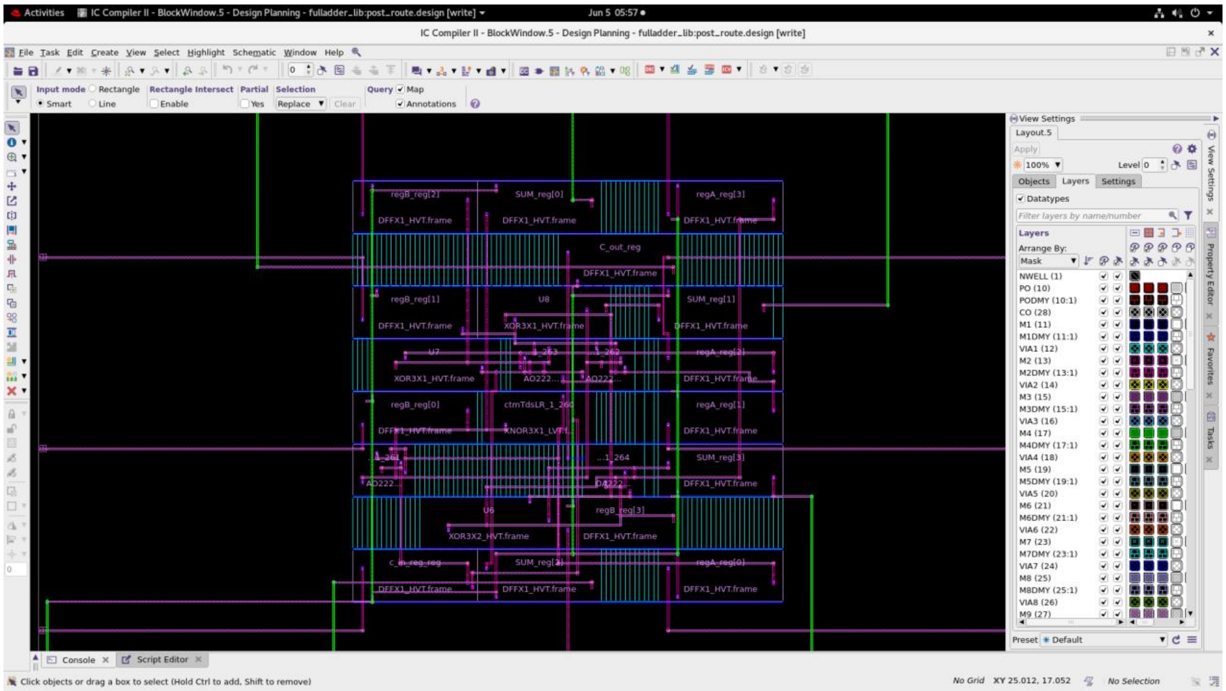

Figure 5.6: Detailed post-route layout view

5.6 Final Routed Design

The final routed design screenshot displays the completed layout in the ICC2 GUI. The view includes standard-cell rows, signal routing, block pins, and the routed design boundary. This screenshot confirms that the implementation reached the post-route design database stage.

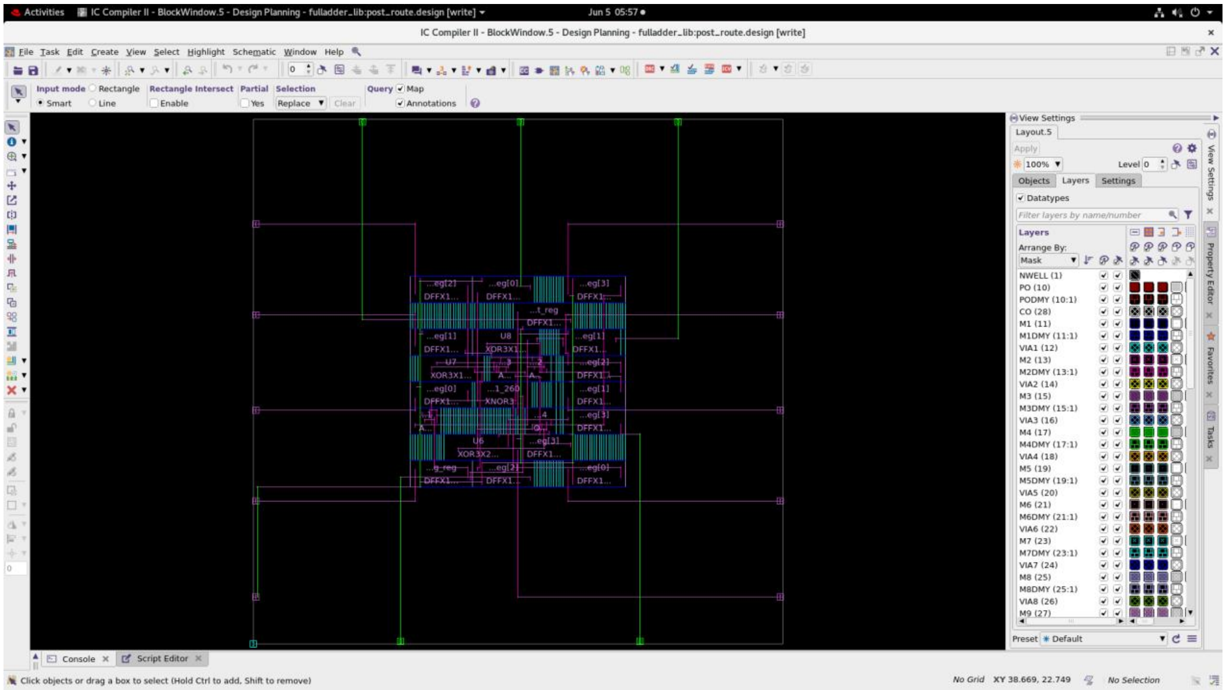

Figure 5.7: Final routed design view

5.7 Filler-cell Insertion

The terminal output reports successful regular filler insertion, including 185 SHFILL1_HVT cells and several larger filler cells. Filler insertion closes empty placement sites and supports continuous well/implant structures. The layout view shows dense filler occupation across the standard-cell rows.

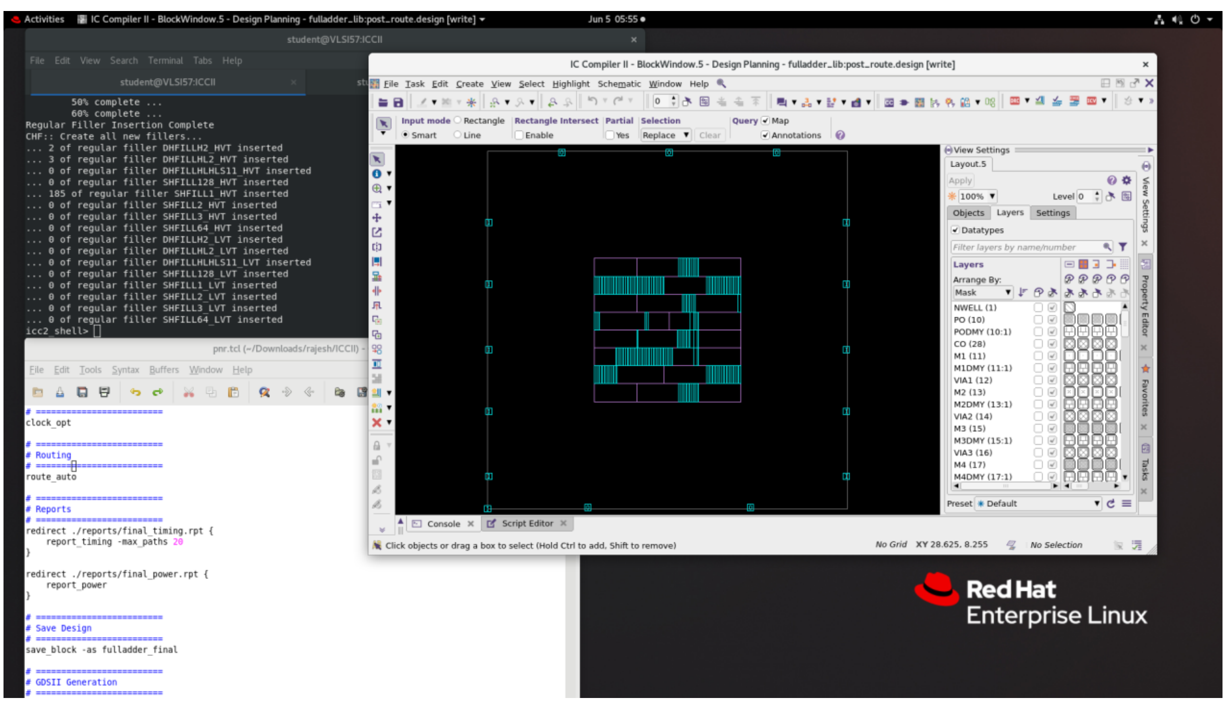

Figure 5.8: Filler-cell insertion and final implementation view

5.8 Script Execution Evidence

The uploaded terminal screenshot records the executed ICC2 sequence, including constraint application, floorplanning, pin placement, power-grid creation, placement stages, clock optimization, routing, filler-cell insertion, GDSII generation, and final checks. User-specific directory names have been removed from the cleaned script documented in this report.

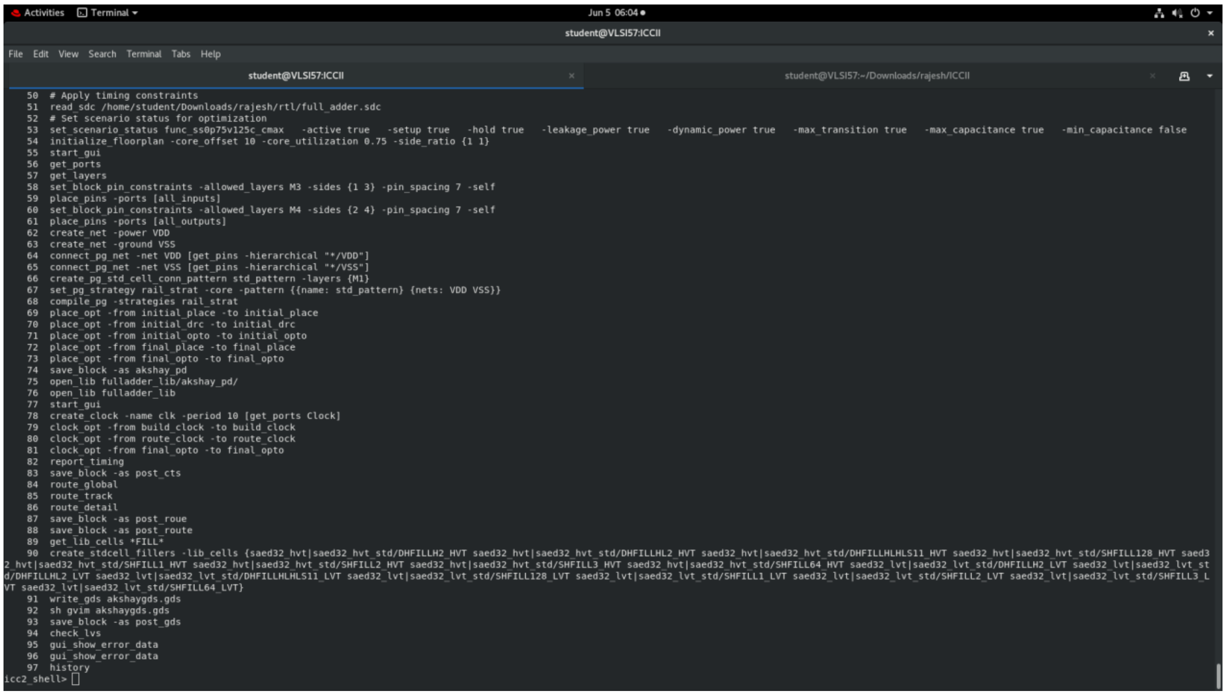

Figure 5.9: ICC2 script and command execution evidence

CHAPTER 6

RESULTS AND ANALYSIS

6.1 Overview of Available Reports

Two sets of synthesis-level area, power, and timing reports were supplied. The earlier state, referred to as the initial reported implementation, contains 69 cells. The later optimized state contains 27 cells. Both reports use the same SAED 32 nm HVT slow-corner library at 0.75 V and 125 degrees Celsius. The reports are compared to show the observed implementation evolution, while differences in clock and activity annotation are explicitly noted.

Metric	Initial report	Optimized report	Observed change
Ports	15	15	No change
Nets	88	38	Reduced by 50
Cells	69	27	60.87% reduction
Combinational cells	55	13	76.36% reduction
Sequential cells	14	14	No change
Buffers/Inverters	22	1	95.45% reduction
Total cell area	236.608068	140.541636	40.60% reduction
Total area	249.617879	147.661879	40.84% reduction
Dynamic power	22.2378 uW	6.4757 uW	70.88% reported reduction
Leakage power	3.0707 uW	2.0678 uW	32.66% reduction
Total power	25.3085 uW	8.5434 uW	66.24% reported reduction

Table 6.1: Comparison of two reported implementation states

6.2 Cell Count Analysis

The optimized report reduces the total standard-cell count from 69 to 27 while preserving all 14 sequential cells. The largest reduction occurs in combinational logic and buffer/inverter cells. This indicates that the later netlist used a more compact logic structure or removed excessive buffering present in the earlier state.

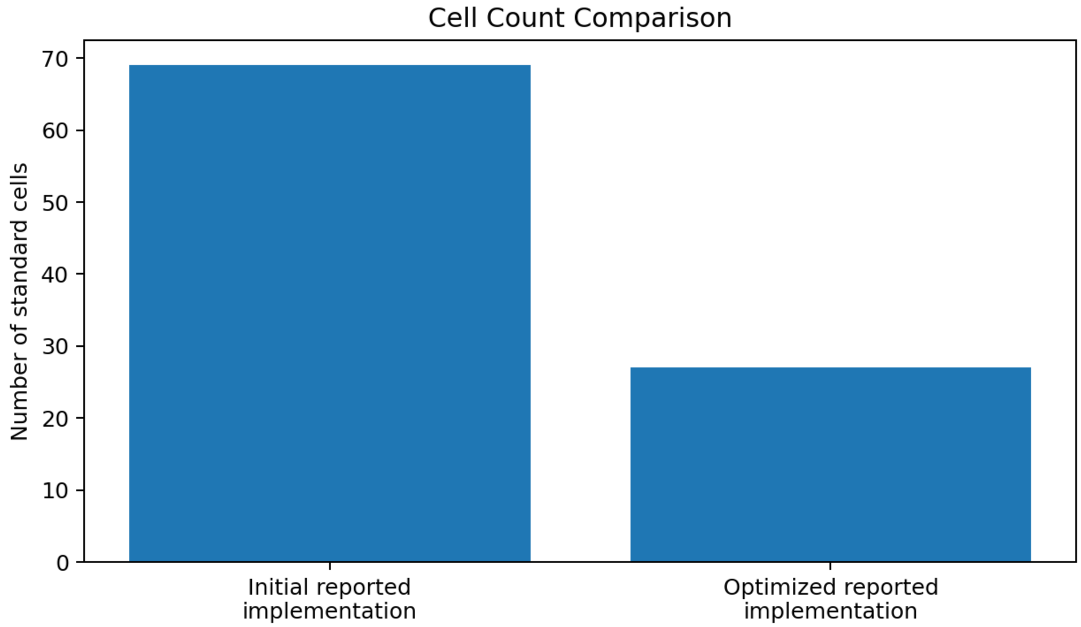

Figure 6.1: Cell-count comparison

6.3 Area Analysis

Area component	Initial report	Optimized report
Combinational area	144.099648	48.033217
Buffer/Inverter area	58.707264	1.270720
Non-combinational area	92.508419	92.508419
Macro/Black-box area	0.000000	0.000000
Estimated interconnect area	13.009812	7.120243
Total cell area	236.608068	140.541636
Total area	249.617879	147.661879

Table 6.2: Detailed area results

The non-combinational area remains constant because the number and type of sequential elements are unchanged. The main area improvement comes from the reduction in combinational logic and buffering. The optimized total area is 147.661879 report units, representing an observed reduction of approximately 40.84% relative to the initial report.

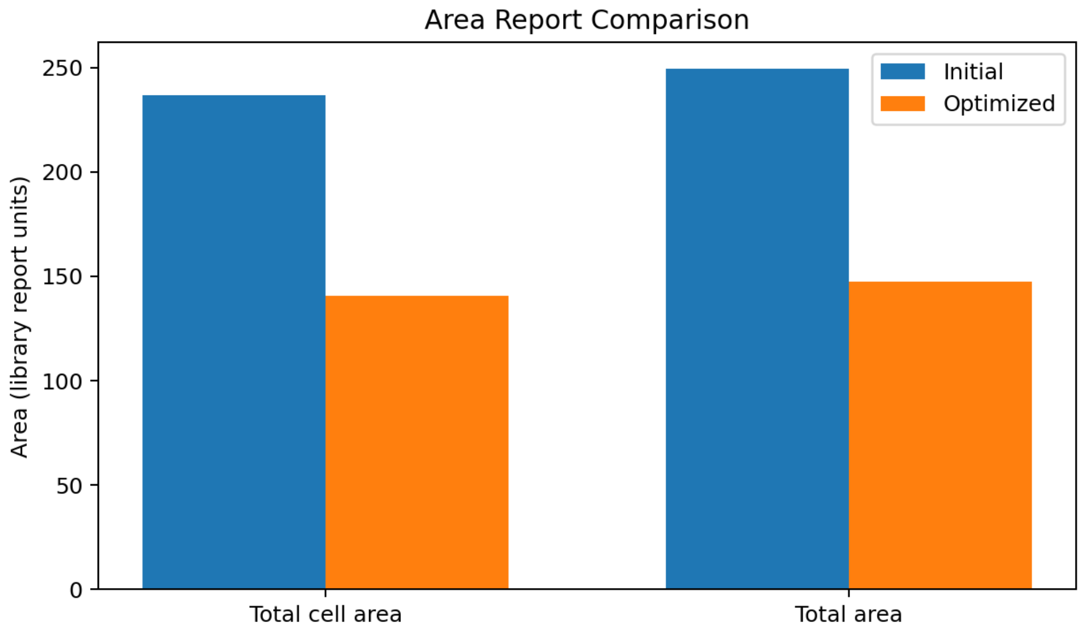

Figure 6.2: Area comparison

6.4 Power Analysis

Power component	Initial report	Optimized report
Cell internal power	20.8399 uW	6.2089 uW
Net switching power	1.3979 uW	0.2667741 uW
Total dynamic power	22.2378 uW	6.4757 uW
Cell leakage power	3.0707 uW	2.0678 uW
Total power	25.3085 uW	8.5434 uW
Clock-network contribution	11.2449 uW	Not included

Table 6.3: Detailed power results

The initial power report includes 11.2449 uW attributed to the clock network and reports a total power of 25.3085 uW. The optimized power report reports 8.5434 uW total power, but it also issues a warning that no clock is defined and states that estimated clock-tree power is not included. Both reports warn that primary-input and sequential-output switching activity is unannotated. The apparent 66.24% total power reduction is therefore an encouraging trend but is not a signoff-quality apples-to-apples comparison.

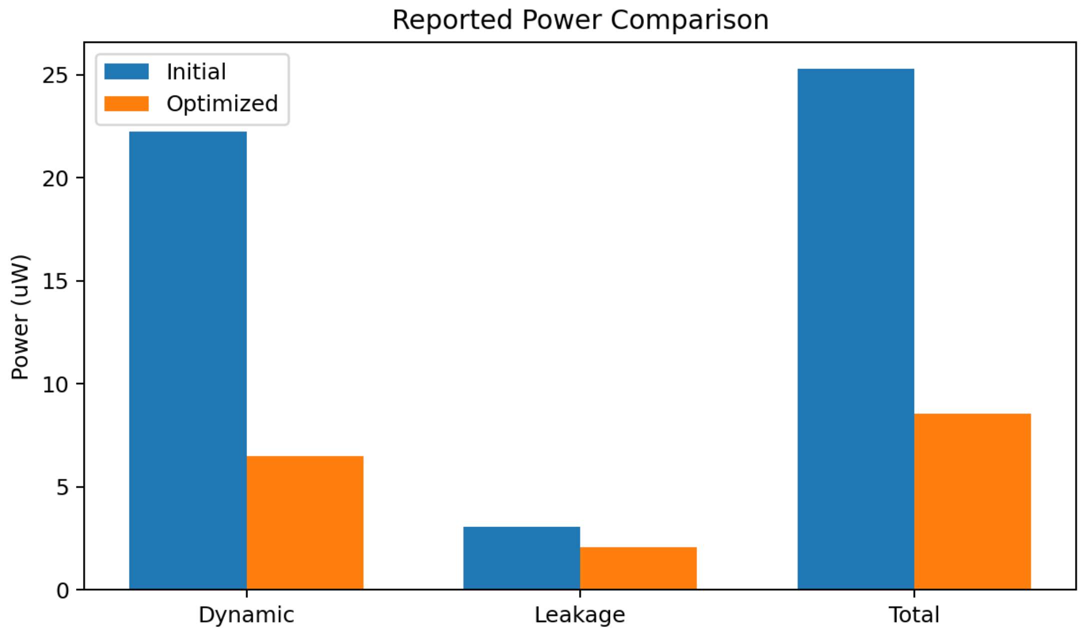

Figure 6.3: Reported power comparison

Power-result limitation

A final power comparison must use the same clock definition, activity annotation, parasitic state, operating scenario, and clock-tree model. The optimized report should be regenerated after reading a complete SDC and applying realistic switching activity, such as SAIF or VCD data.

6.5 Timing Analysis

Timing item	Report value
Initial critical path	regB_reg[0] to SUM_reg[3]
Path type	Maximum-delay register-to-register path
Data arrival time	6.90 ns
Data required time	2.72 ns
Slack	-4.19 ns (violated)
Final reported path	SUM_reg[3] to output SUM[3]
Final path delay	1.95 ns
Final path status	Unconstrained
Physical-design clock	10 ns period applied in ICC2 script

Table 6.4: Timing report interpretation

The initial report shows a setup violation of -4.19 ns on a register-to-register path. The path contains several complex gates and buffers and arrives at 6.90 ns, while the required time is 2.72 ns. The final report shows a shorter 1.95 ns clock-to-output path, but explicitly marks it unconstrained. Therefore, the final timing report cannot be used to claim timing closure. A complete final SDC and post-route timing report are required to establish setup and hold closure at the intended 10 ns clock period.

6.6 Floorplan and Placement Analysis

The final rectangular floorplan uses 75% core utilization and a core offset of 10. The uploaded floorplan report states that no floorplan rules exist, which means no separate rule set was loaded; it does not by itself indicate an invalid floorplan. The small design is concentrated in the center of the core, and the screenshots show legal cell rows and boundary pin placement. The generous outer whitespace reduces density pressure but creates long routes from the central logic to some boundary pins.

6.7 Power-grid Verification

Verification metric	Result
Standard cells checked	27
VDD wires	6
Floating VDD wires	5
Floating standard cells on VDD	27
VSS wires	5
Floating VSS wires	4
Floating standard cells on VSS	27
PG DRC	No errors found

Table 6.5: Power-grid verification results

The power-grid DRC report contains no geometrical errors, indicating that the generated PG shapes do not violate the checked width, spacing, or enclosure rules. The connectivity report, however, identifies floating VDD and VSS wires and reports all 27 standard cells as floating on both supplies. This means the power-grid geometry is present but not electrically connected to the cell rails in the analyzed database. The design must not be described as power-grid signoff clean until this issue is repaired.

6.8 Filler Insertion and GDSII

The filler insertion log shows that regular fillers were inserted successfully. The most numerous filler was SHFILL1_HVT, with 185 instances, together with several larger HVT filler cells. The command history also shows GDSII generation and a post-GDS saved block. These steps confirm completion of the physical flow demonstration, although final fabrication readiness remains dependent on clean connectivity, complete timing constraints, route checks, and formal signoff verification.

CHAPTER 7

VERIFICATION FINDINGS AND CORRECTIVE ACTIONS

7.1 Verification Summary

Flow item	Status	Evidence / observation
Technology/library setup	Completed	SAED 32 nm HVT/LVT libraries and TLU+ models loaded.
Netlist import and linking	Completed	15-port sequential adder imported and linked.
MMMC scenario	Completed	Functional slow corner configured at 0.75 V and 125 C.
Floorplanning	Completed	Rectangular 75% utilization floorplan documented.
Placement	Completed	Cells placed and visible in legal rows.
Clock-tree synthesis	Executed	10 ns clock created in ICC2 script.
Signal routing	Completed visually	Post-route and final layout screenshots available.
Filler insertion	Completed	Filler insertion log available.
PG geometrical DRC	Clean	No errors found.
PG connectivity	Not clean	All 27 cells reported floating on VDD and VSS.
Final timing closure	Not demonstrated	Final timing path is unconstrained.
Power signoff	Not demonstrated	Clock and switching activity are incompletely annotated.
GDSII generation	Completed	GDS write command and saved post-GDS block documented.

Table 7.1: Signoff status and required corrective actions

7.2 Power-grid Connectivity Repair

1. Confirm the actual power and ground pin names in both HVT and LVT standard-cell libraries.
2. Use connect_pg_net -automatic or explicit UPF-based supply-set connectivity where supported.
3. Verify that standard-cell rows contain M1 rails aligned with the cell power pins.
4. Create sufficient vias or via ladders between the rails and higher-level straps or rings.
5. Add top-level VDD and VSS terminals if required by the flow.
6. Run check_pg_connectivity after each PG change and inspect reported floating objects in the GUI.
7. Proceed to IR-drop analysis only after the connectivity report contains no floating standard cells.

7.3 Timing-constraint Repair

The final SDC must contain the clock and external timing environment rather than relying on an interactive create_clock command executed later in the flow. A suitable starting point is shown below. The actual delays and loads should be selected from system-level requirements and library limits.

```
set_units -time ns -capacitance fF -voltage V
create_clock -name clk -period 10.0 [get_ports Clock]
set_clock_uncertainty 0.15 [get_clocks clk]
set_input_delay  1.0 -clock clk [remove_from_collection [all_inputs] [get_ports Clock]]
set_output_delay 1.0 -clock clk [all_outputs]
set_input_transition 0.10 [all_inputs]
set_load 5.0 [all_outputs]
set_max_transition 0.50 [current_design]
set_max_fanout 10 [current_design]
```

7.4 Pre-place Check Correction

The uploaded pre-place check report shows that the requested check name pre_place_opt_stage is not valid. This is a command-usage error rather than a design-rule result. The corrected approach is to run check_design without the invalid check name, query the supported checks for the installed ICC2 version, or invoke the documented pre-placement check set. The result should then be redirected to a report file and reviewed before place_opt.

7.5 Floorplan and Routing Improvements

- Reduce unnecessary die-to-core whitespace for this small block to shorten boundary routes.
- Place related input pins closer to the operand register banks and output pins closer to the result registers.
- Retain a single floorplan initialization in the final script to avoid overwriting earlier geometry.
- Use route congestion and QoR reports to determine whether pin layers M3/M4 improve access compared with M1/M2.
- Run route_opt after detailed routing and regenerate setup, hold, transition, capacitance, and route-DRC reports.
- Use explicit approved filler-cell lists when automatic wildcard matching returns non-filler cells.

7.6 Recommended Final Signoff Checklist

No.	Final signoff requirement
1	All clocks and IO constraints are present in the final SDC.
2	No unconstrained timing paths remain.
3	Setup WNS and TNS are non-negative or formally waived.
4	Hold WNS and TNS are non-negative or formally waived.
5	Maximum transition, capacitance, and fanout constraints are clean.
6	All signal routes are complete with no shorts or opens.
7	PG connectivity reports zero floating cells, wires, vias, and terminals.
8	PG DRC is clean.

No.	Final signoff requirement
9	Clock skew, latency, and transition satisfy the project target.
10	Power is regenerated with clock-tree power and realistic switching activity.
11	Final filler, well-tap, boundary, and antenna requirements are satisfied.
12	GDSII, netlist, SDC, SPEF, and final reports are archived consistently.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project successfully demonstrates the major stages of an ASIC physical-design flow for a 4-bit sequential adder using Synopsys ICC2. The team configured the SAED 32 nm technology environment, imported a synthesized design containing 14 sequential cells, created floorplans, placed block pins, constructed a standard-cell power structure, performed multi-stage placement, created a 10 ns clock, executed clock optimization, completed signal routing, inserted filler cells, generated reports, and exported the final layout to GDSII.

The optimized synthesis report contains 27 cells and a total area of 147.661879 report units. Relative to the earlier 69-cell report, it shows substantial reductions in cell count, buffer count, area, and reported power. The physical screenshots provide clear evidence of placement, routing, post-route inspection, filler insertion, and final layout generation.

The analysis also identifies two important signoff limitations. First, the final uploaded timing report is unconstrained because the supplied SDC does not contain a clock or IO timing environment. Second, the power-grid connectivity report identifies all standard cells as floating even though power-grid geometrical DRC is clean. The project therefore achieves its educational objective of implementing and studying the complete physical flow, while also demonstrating why report-based verification is essential before declaring a design timing-clean or fabrication-ready.

8.2 Skills Acquired

- ASIC RTL-to-GDSII flow understanding
- Synopsys ICC2 design-library and block management
- SAED 32 nm technology and standard-cell library setup
- MMMC mode, corner, and scenario configuration
- TLU+ parasitic model setup
- Floorplanning and pin placement
- Power-grid strategy creation and verification
- Timing-driven placement and legalization
- Clock Tree Synthesis and clock optimization
- Global, track, and detailed routing
- Area, power, timing, and connectivity report interpretation
- Filler-cell insertion and GDSII generation
- TCL scripting and reproducible flow organization
- Identification of incomplete constraints and signoff issues

8.3 Future Scope

- Repair the VDD/VSS connectivity and rerun all PG checks.

- Create a complete SDC and close both setup and hold timing at 100 MHz.
- Perform post-route extraction and generate SPEF and SDF outputs.
- Annotate realistic switching activity and repeat power analysis.
- Run IR-drop and electromigration analysis on the completed power grid.
- Compare rectangular and L-shaped floorplans using area, wirelength, congestion, and timing metrics.
- Compare HVT-only, LVT-only, and mixed-Vt implementations.
- Extend the arithmetic block to an 8-bit or 16-bit sequential adder.
- Automate the full flow through a validated master TCL script and stage-wise error checks.
- Perform formal equivalence checking between the synthesized and final implementation netlists.

CHAPTER A

CLEANED ICC2 COMMAND FLOW

The following compact script consolidates the uploaded command history into a readable stage-by-stage flow. All user names, personal directories, and unrelated institution names have been removed. Paths must be replaced with valid local technology, library, netlist, and constraint locations.

```
# 1. Create project folders
file mkdir ./reports
file mkdir ./results
file mkdir ./logs

# 2. Set project variables
set DESIGN_NAME      "your_top_module"
set DESIGN_LIBRARY   "${DESIGN_NAME}_lib"
set TECH_FILE        "/path/to/technology.tf"
set NETLIST_FILE     "/path/to/design_netlist.v"
set SDC_FILE         "/path/to/design_constraints.sdc"
set REF_LIBS         [list "/path/to/hvt.ndm" "/path/to/lvt.ndm"]
set DB_LIBS          [list "/path/to/hvt.db" "/path/to/lvt.db"]
set MAX_TLU          "/path/to/max.tluplus"
set MIN_TLU          "/path/to/min.tluplus"
set TLU_MAP          "/path/to/layer.map"

# 3. Create and open the design library
if {[file exists $DESIGN_LIBRARY]} { file delete -force $DESIGN_LIBRARY }
create_lib -technology $TECH_FILE -ref_libs $REF_LIBS $DESIGN_LIBRARY
open_lib $DESIGN_LIBRARY
set_app_var link_library   [concat {*} $DB_LIBS]
set_app_var target_library $DB_LIBS
read_parasitic_tech -name maxTLU -tlup $MAX_TLU -layermap $TLU_MAP
read_parasitic_tech -name minTLU -tlup $MIN_TLU -layermap $TLU_MAP

# 4. Import and link the gate-level netlist
read_verilog $NETLIST_FILE -library $DESIGN_LIBRARY     -design "${DESIGN_NAME}_net" -top $DESIGN_NAME
current_design $DESIGN_NAME
link_block
check_design

# 5. Create mode, corner, and scenario
remove_scenarios -all
remove_modes -all
remove_corners -all
create_mode functional
create_corner slow_max
set_voltage -object_list {VDD} 0.75
set_voltage -object_list {VSS} 0.0
set_temperature -corners slow_max 125
set_parasitic_parameters -corners slow_max     -late_spec maxTLU -early_spec minTLU
create_scenario -name functional_slow_max     -mode functional -corner slow_max
current_mode functional
current_corner slow_max
current_scenario functional_slow_max
read_sdc $SDC_FILE
set_scenario_status functional_slow_max     -active true -setup true -hold true     -leakage_power true -dynamic_power true     -max_transition
true -max_capacitance true

# 6. Floorplan and pins
initialize_floorplan -shape R -core_utilization 0.75     -side_ratio {1 1} -core_offset 10
set_block_pin_constraints -allowed_layers M3     -sides {1 3} -pin_spacing 7 -self
place_pins -ports [all_inputs]
set_block_pin_constraints -allowed_layers M4     -sides {2 4} -pin_spacing 7 -self
place_pins -ports [all_outputs]
save_block -as floorplan

# 7. Power planning
```

```
create_net -power VDD
create_net -ground VSS
connect_pg_net -net VDD [get_pins -hierarchical "*/VDD"]
connect_pg_net -net VSS [get_pins -hierarchical "*/VSS"]
create_pg_std_cell_conn_pattern rail_pattern -layers {M1}
set_pg_strategy rail_strategy -core     -pattern {{name: rail_pattern} {nets: VDD VSS}}
compile_pg -strategies rail_strategy
check_pg_connectivity
check_pg_opens
check_pg_drc
save_block -as power_plan

# 8. Placement
place_opt -from initial_place -to initial_place
place_opt -from initial_drc   -to initial_drc
place_opt -from initial_opto  -to initial_opto
place_opt -from final_place   -to final_place
place_opt -from final_opto    -to final_opto
save_block -as post_placement

# 9. Clock Tree Synthesis
if {[sizeof_collection [get_clocks -quiet clk]] == 0} {
    create_clock -name clk -period 10 [get_ports Clock]
}
clock_opt -from build_clock -to build_clock
clock_opt -from route_clock -to route_clock
clock_opt -from final_opto  -to final_opto
report_timing > ./reports/post_cts_timing.rpt
save_block -as post_cts

# 10. Routing
route_global
route_track
route_detail
route_opt
check_routes
save_block -as post_route

# 11. Filler insertion and final checks
set FILLER_CELLS [get_lib_cells -quiet */*FILL*]
if {[sizeof_collection $FILLER_CELLS] > 0} {
    create_stdcell_fillers -lib_cells $FILLER_CELLS
}
check_legality
check_routes
check_pg_connectivity
check_pg_drc
check_timing
report_qor > ./reports/final_qor.rpt
report_timing -delay_type max > ./reports/final_setup.rpt
report_timing -delay_type min > ./reports/final_hold.rpt
save_block -as signoff_ready

# 12. GDSII export
write_gds ./results/final_layout.gds
save_block -as post_gds
```

CHAPTER B

SUMMARY OF UPLOADED REPORTS

B.1 Optimized Area Report

```
Design: full_adder
Ports: 15
Nets: 38
Cells: 27
Combinational cells: 13
Sequential cells: 14
Buffers/Inverters: 1
Combinational area: 48.033217
Non-combinational area: 92.508419
Total cell area: 140.541636
Total area: 147.661879
```

B.2 Optimized Power Report

```
Operating voltage: 0.75 V
Cell internal power: 6.2089 uW
Net switching power: 0.2667741 uW
Total dynamic power: 6.4757 uW
Cell leakage power: 2.0678 uW
Total power: 8.5434 uW
Warning: No defined clock; estimated clock-tree power not included.
Warning: Primary inputs and sequential outputs are unannotated.
```

B.3 Initial Timing Report

```
Startpoint: regB_reg[0]
Endpoint: SUM_reg[3]
Path type: Maximum delay
Data arrival time: 6.90 ns
Data required time: 2.72 ns
Slack: -4.19 ns (VIOLATED)
```

B.4 Optimized Timing Report

```
Startpoint: SUM_reg[3]
Endpoint: SUM[3]
Path delay: 1.95 ns
Path status: Unconstrained
```

B.5 Power-grid Reports

```
PG DRC: No errors found.
Standard cells checked: 27
VDD wires: 6; floating VDD wires: 5
VSS wires: 5; floating VSS wires: 4
Floating standard cells on VDD: 27
Floating standard cells on VSS: 27
```

B.6 Supplied SDC Content

```
set sdc_version 2.1
set_units -time ns -resistance MOhm     -capacitance fF -voltage V -current uA

# No clock or IO timing constraints were present in the supplied file.
```

REFERENCES

[1] Synopsys, IC Compiler II documentation and command reference.

[2] Synopsys, Static Timing Analysis and SDC constraint documentation.

[3] SAED 32 nm standard-cell and technology-library documentation.

[4] Neil H. E. Weste and David Harris, CMOS VLSI Design: A Circuits and Systems Perspective.

[5] Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolic, Digital Integrated Circuits: A Design Perspective.